

Quantitative Trading Strategy Development on NSE 500 Universe



Hostel
2601

555750	9	15,040 €	15,041 €	15,041 €	1,35%
833212	10	12,648 €	12,648 €	12,648 €	1,35%
843002	3	176,020 €	176,004 €	176,042 €	1,35%
510000	23	88,864 €	88,870 €	88,871 €	1,35%

Problem Statement & Core Objectives

- Long-only equity strategy on a universe of ~900 NSE stocks.
- Backtesting over January 2010 to December 2020.
- Consistent risk-adjusted out performance over the Nifty 500 benchmark.
- Disciplined investement through stock selection, weight allocation, and risk mitigation.
- Performance is evaluated primarily by the Sharpe Ratio and CAGR, making risk-adjusted return maximization the ultimate objective.
- Execution price computed as OHLC mean:

$$P_{execu} = \frac{O + H + L + C}{4}$$

Static Weights Fail

A static MVO portfolio constructed once at the start and held without re-balancing drifts from its optimal composition as market conditions evolve.

Exploring Market Dynamics

Noisy Universe

With ~900 stocks across sectors, high-dimensionality made raw optimization prone to noise. Hence, making stock selection necessary.

Regime Shifts

The data showed clear structural shifts — bull phases (2014–2018), sector rotation events, and risk-off episodes (2018–2019 trade war). Correctly identifying the market regime proved critical.

Large Drawdowns

Nifty experiences large drawdowns leading to significant losses and slow recovery, highlighting the need for risk management like stop-losses and diversification.



The diagram features a central purple circle with the text "Experimented Approaches" in white. This circle is surrounded by a larger, faint white circle. To the right of the central circle, five white rounded rectangular boxes are stacked vertically, each containing a different stock selection approach. A thin white line connects the central circle to the first box, and a larger, faint white circle encompasses the entire set of boxes. The background is dark with a faint, stylized candlestick chart in red and teal.

Experimented Approaches

Mean Variance Optimization (Multiple Objectives)

MVO With Rebalancing (Static + Volume Triggered)

Score Based Stock Selection(VAM,Trend,Liquidity)

Sectoral Relative Strength Analysis

Residual Momentum Stock Selection

Mean Variance Optimization

Three canonical objectives were examined:

- Maximum Sharpe Ratio
- Minimum Volatility
- Maximum Return

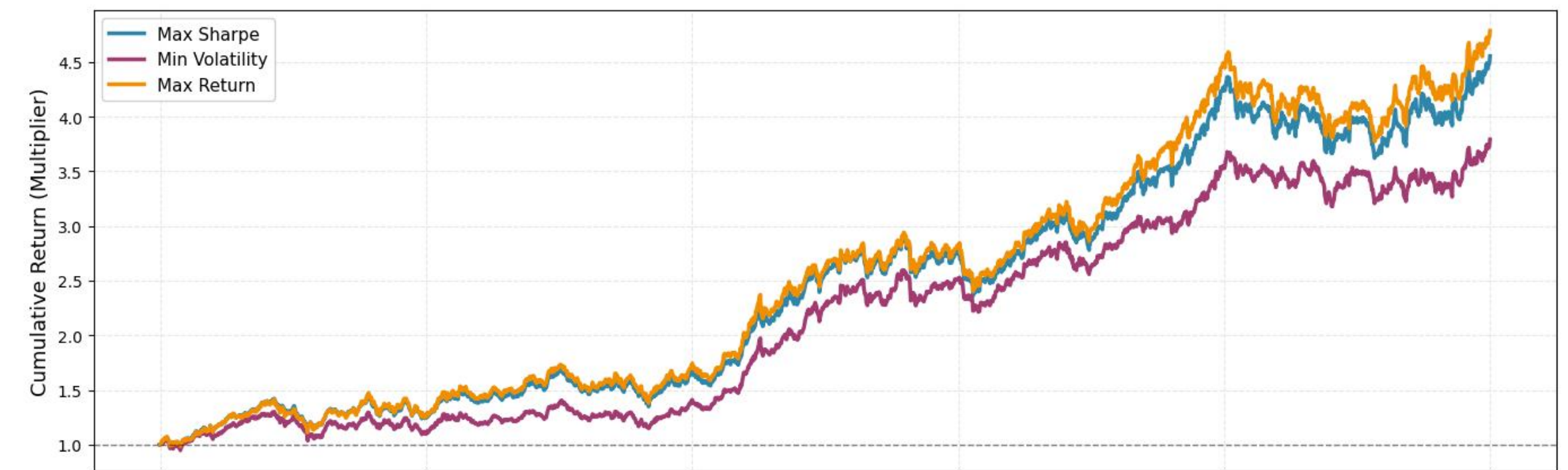
Solved using SLSQP with long-only constraints over the full universe, 252-day lookback period to construct return and covariance matrix.

Max Sharpe : $\min_w - \frac{\mu^\top w}{\sqrt{w^\top \Sigma w}}$ Min Volatility: $\min_w w^\top \Sigma w$

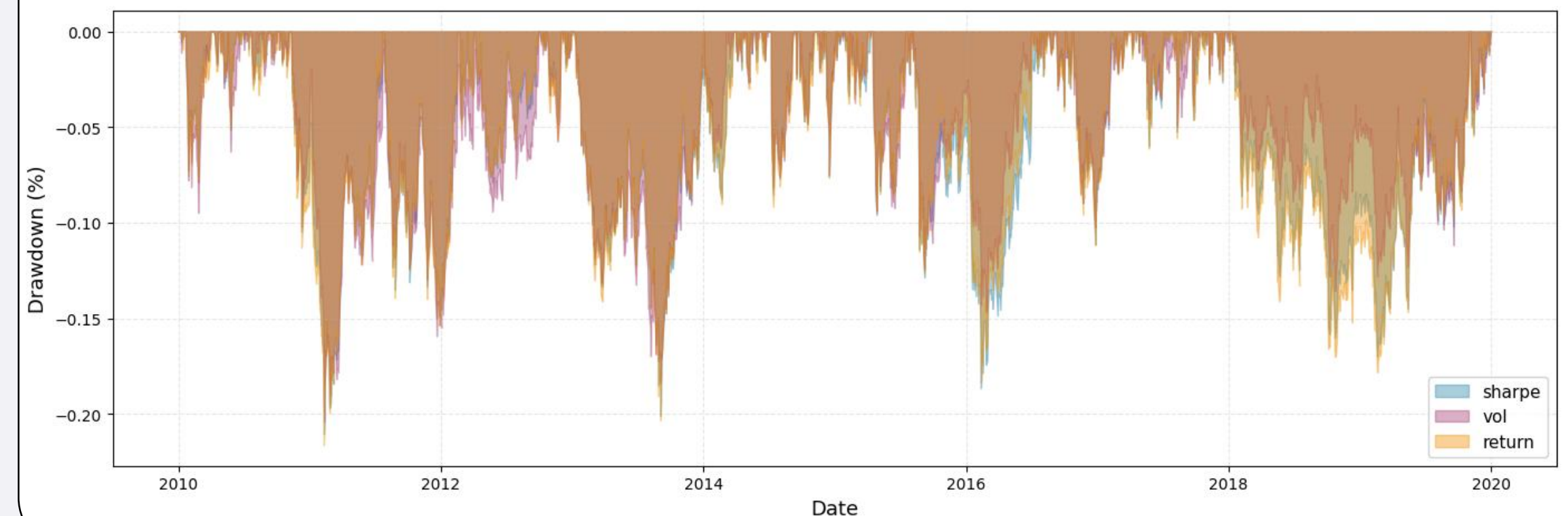
Objective	Sharpe	CAGR	Max DD
Max Sharpe	1.20	16.66%	-21.06%
Min Vol	1.12	14.50%	-20.40%
Max Return	1.19	17.24%	-21.63%

Equity and drawdown plots for comparison

Portfolio Growth Comparison (2010 - 2020)



Portfolio Drawdowns



MVO with Rebalancing

To keep up with the evolving market we need to update our allocations periodically.

Two types of rebalancing triggers were tested:

- **Static 30-day rebalancing**
- **Volume-Spike triggered rebalancing:** Whenever the volume crossed the threshold $\theta = 1.35 \times 60$ -day baseline.

Volume spike + 30 day rebalancing clearly outperforms buy and hold !

Volume spikes often precede significant price moves; early repositioning captures momentum or limits drawdown.



Objective	Sharpe	CAGR	Max DD
Buy and Hold	1.13	15.10%	-17.37%
Static (30 day)	1.31	17.92%	-21.49%
Volume + Static	1.42	19.35%	-21.15%

Score Based Selection (VAM, Trend and Liquidity)

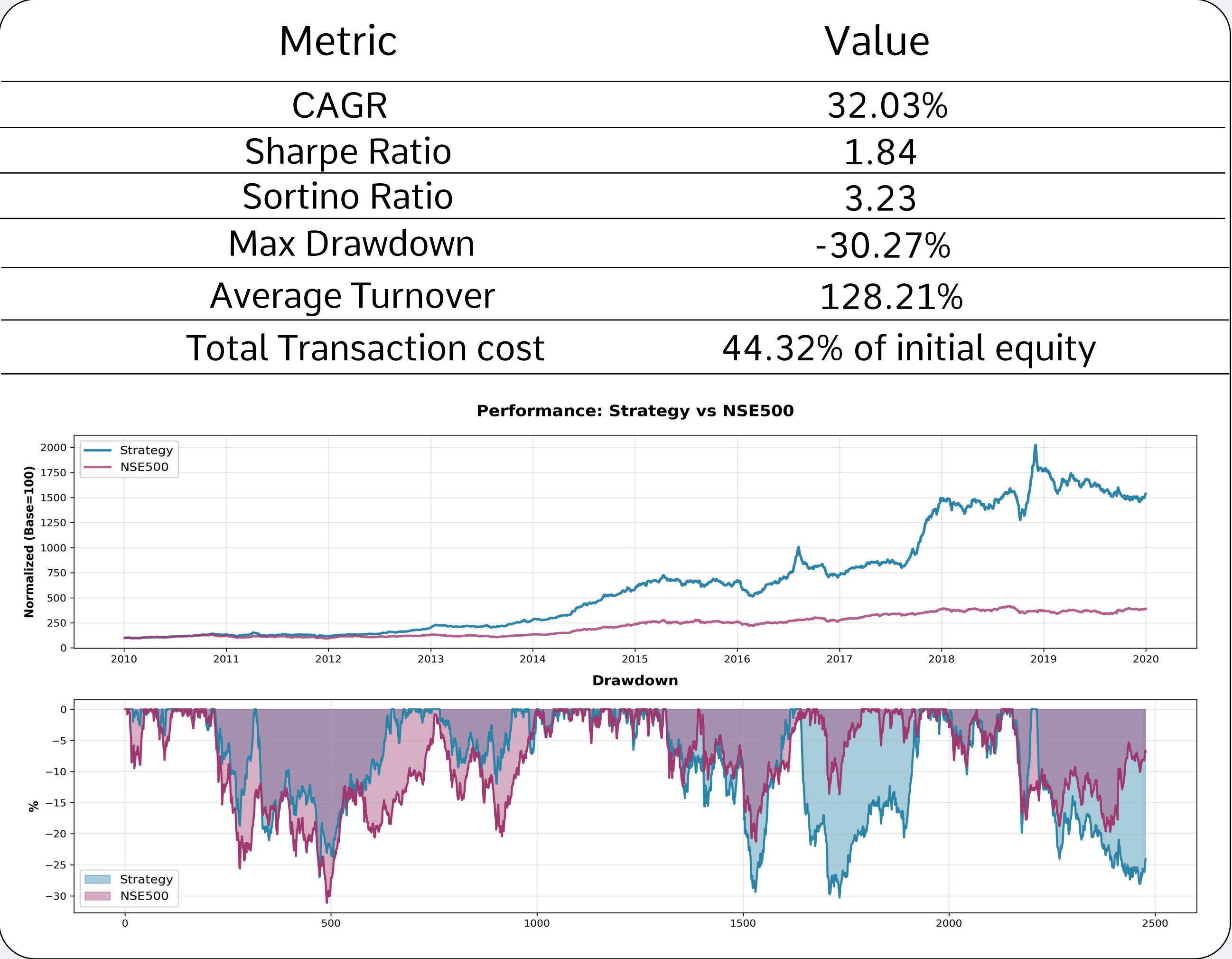
To filter out only the performing stocks we tried a scoring based approach. Top 100 Stocks are selected after calculating a score, followed by MVO Sharpe for weight allocation.

$$Q_t = 0.5 \cdot \text{VAM} + 5.0 \cdot \text{Trend} - 2.0 \cdot \text{Liq}$$

- Volatility adjusted momentum: $VAM_t = \frac{r(20)_t}{\sigma(20)_t}$
- Trend score: $Trend_t = \frac{C_t - \text{SMA20}}{\text{SMA20}}$
- Liquidity score:
 $Liq_t = \text{rolling 20-day avg of } \left(\frac{\text{Traded Value}}{\text{MCap}} \right)$

Limitations

- **High Turnover:** 128.21% average turnover implies excessive transaction costs.
- **Risk of parameter overfitting:** The composite score used are somewhat arbitrarily chosen weights, introducing a risk of parameter overfitting.



Relative Strength Analysis

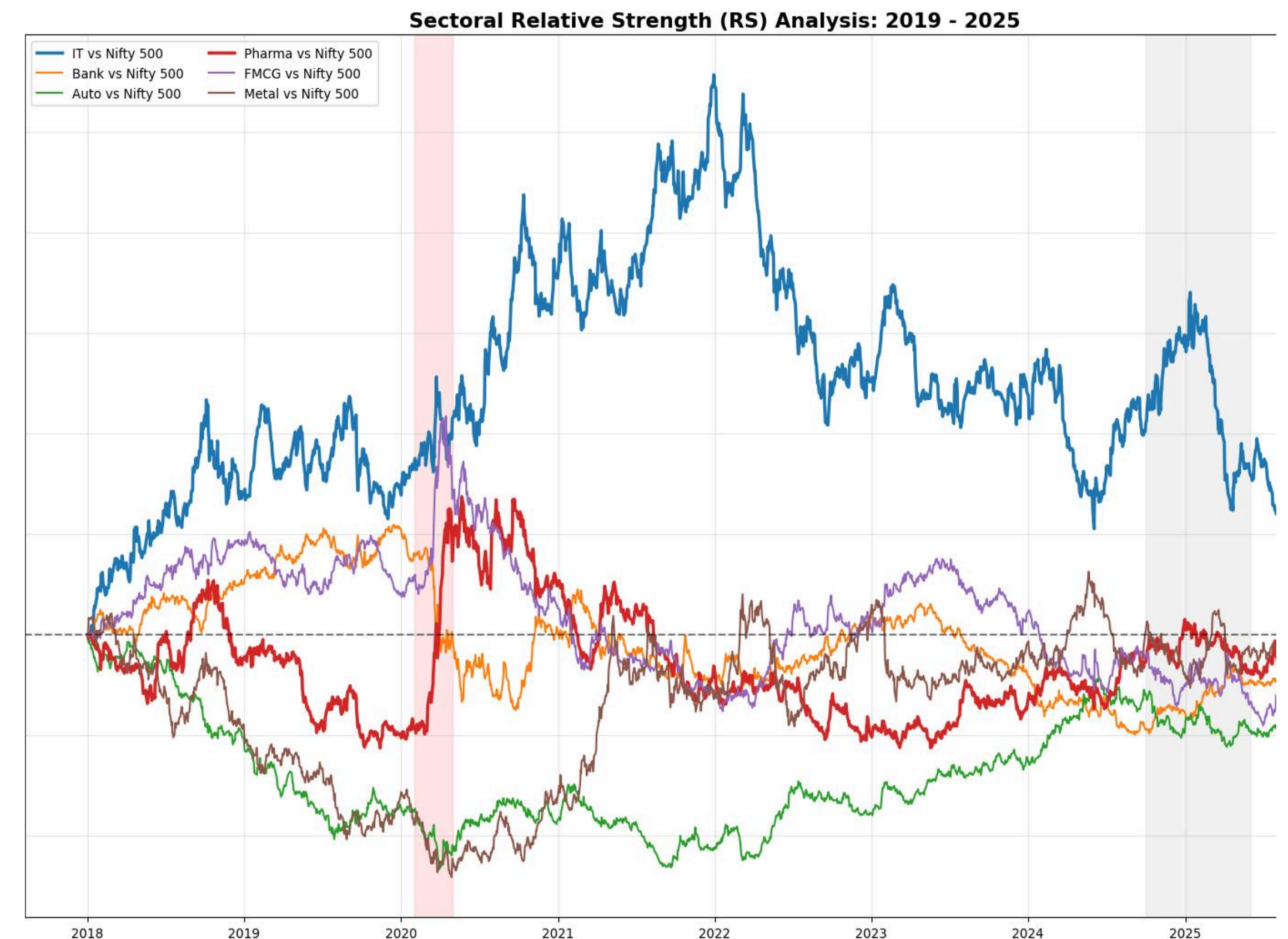
RS ratio measures each sector's or stock's performance relative to the Nifty 500 benchmark.

RS > 1.0 = outperforming.

The strategy selects top-K stocks by RS as a macro-level pre-filter before scoring, ensuring capital is only allocated to outperforming stocks

$$RS_{s,t} = \frac{P_{s,t}/P_{b,t}}{P_{s,0}/P_{b,0}}$$

- Pharma dominated 2010-2015, then structural decline (regulatory headwinds).
- IT and Banking rotated sharply around the 2014 General Election.
- Metals was the most consistent underperformer.
- FMCG served as a steady defensive anchor.

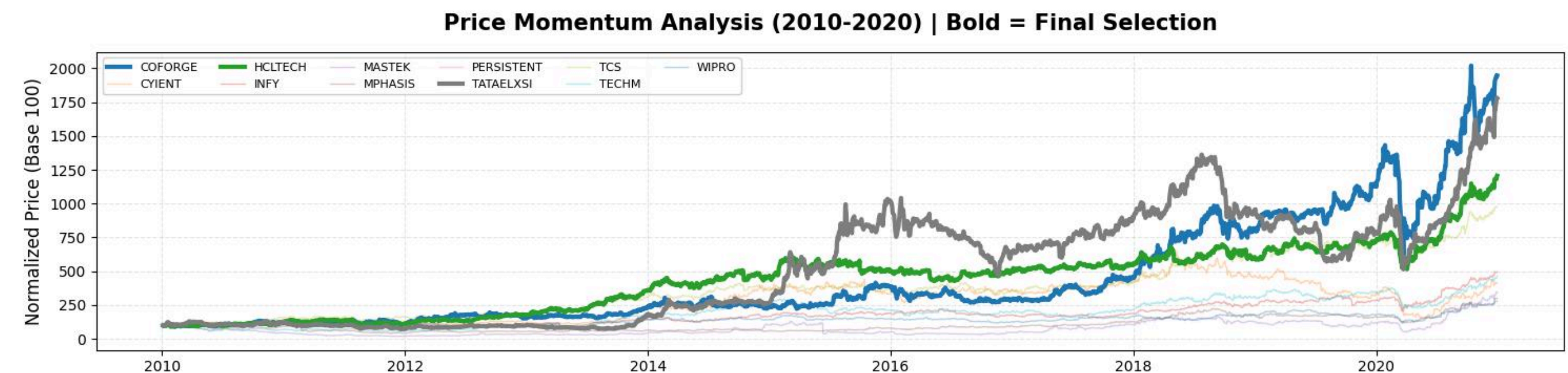


Residual Momentum Stock Selection

We tested an unique and simple scoring technique, in which we treat the stock price as data points and fit a line on these points. The fitted line gives us two things:

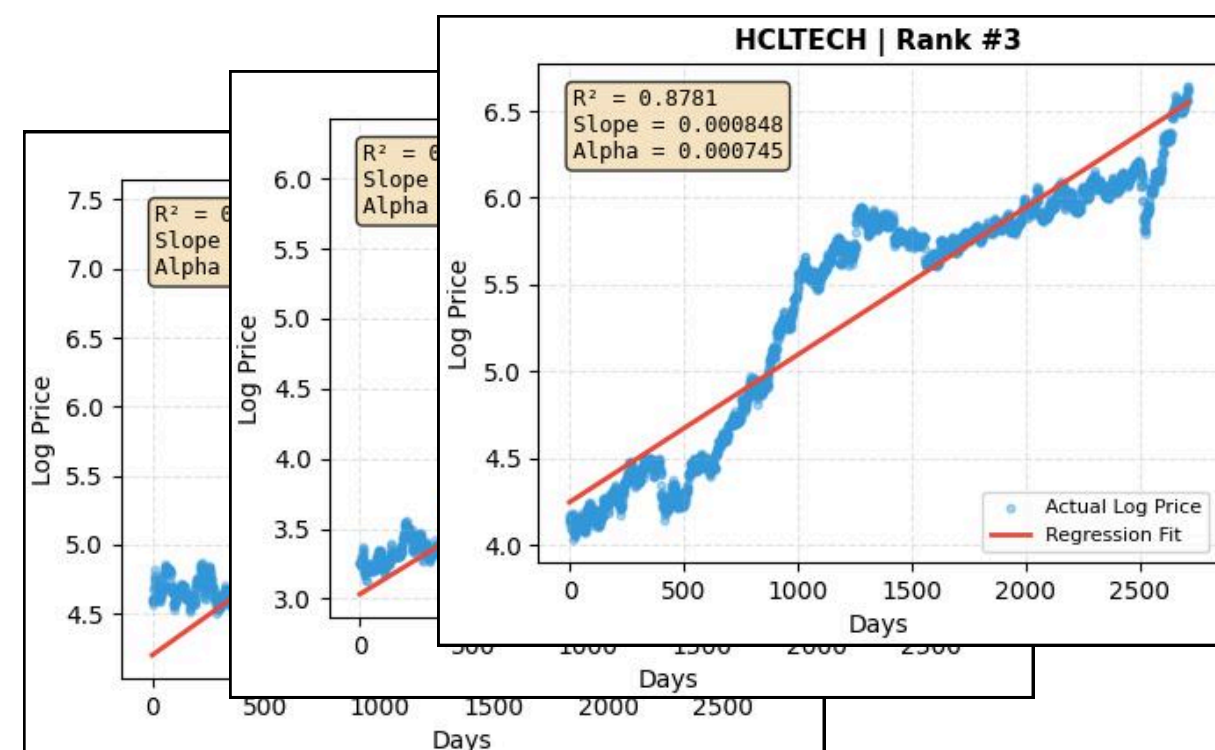
1. **Slope:** Signifying trend strength.
2. **Residual:** Signifying trend consistency

The selected stocks clearly outperform their counterparts !!



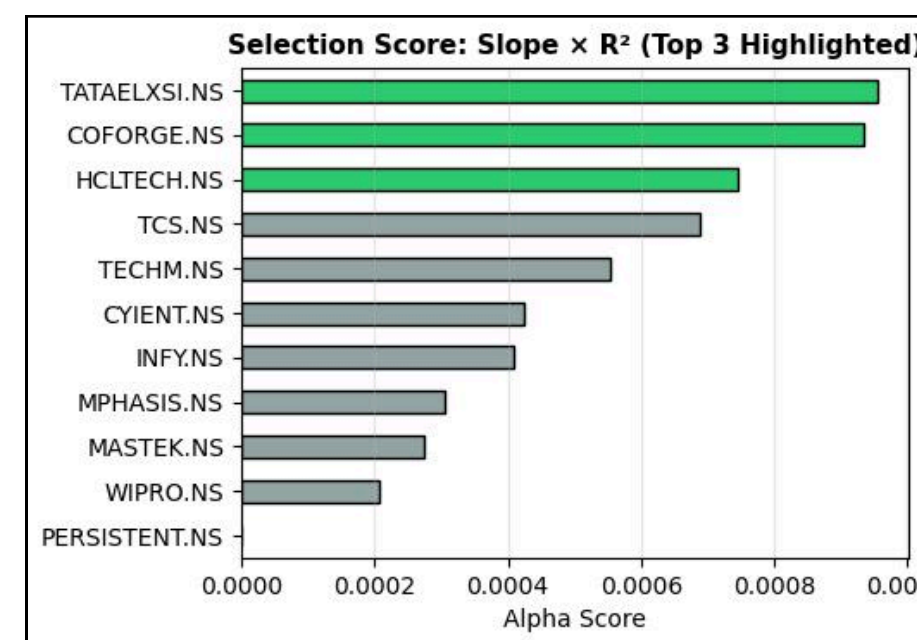
Step 1: Log-Linear Regression on each stock's log price with time.

$$\log P_{i,t} = \alpha_i + \beta_i \cdot t + \epsilon_{i,t}$$

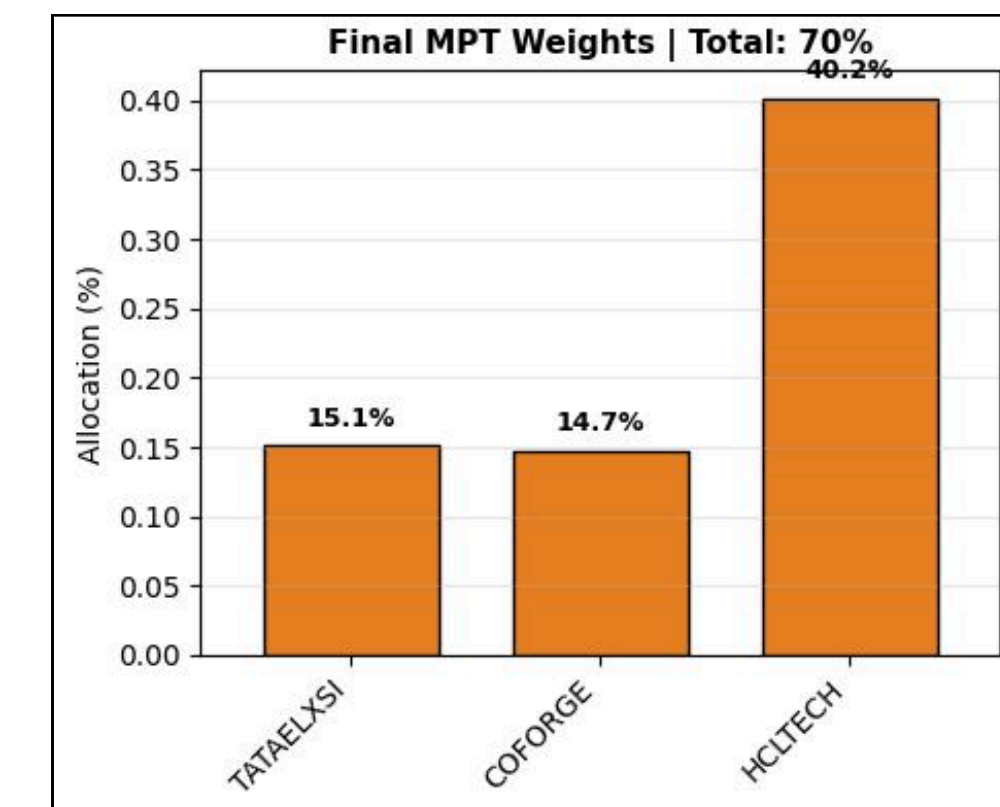


Step 2: Calculate the score for each stock using the slope and residuals.

$$\alpha_i = \beta_i \cdot R_i^2 \quad R_i^2 = 1 - \frac{\sum \epsilon^2}{\sum (\log P - \log P)^2}$$

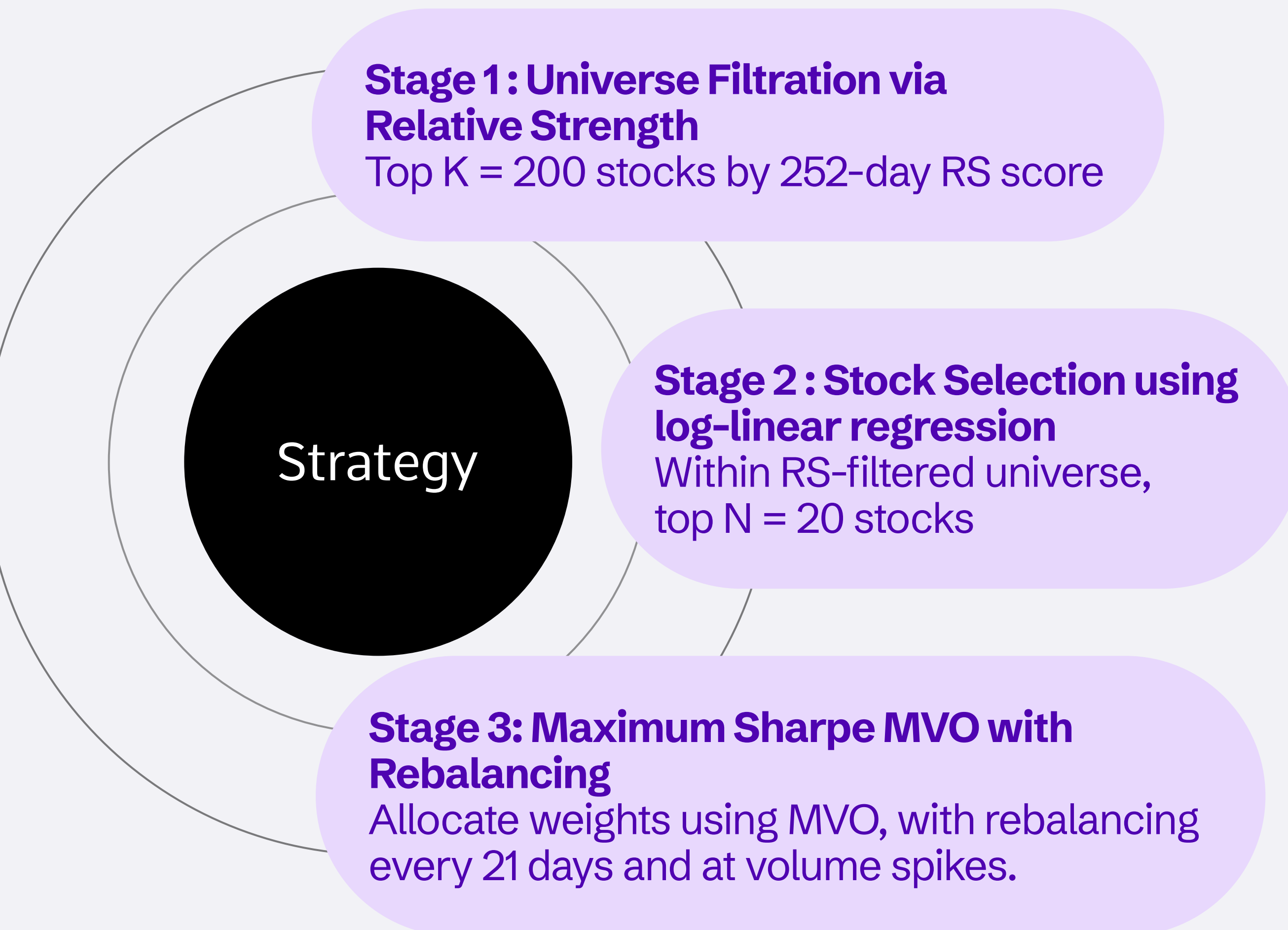


Step 3: Allocate weights to the selected stocks using MVO.



Our Final Approach:

Three-Stage Pipeline



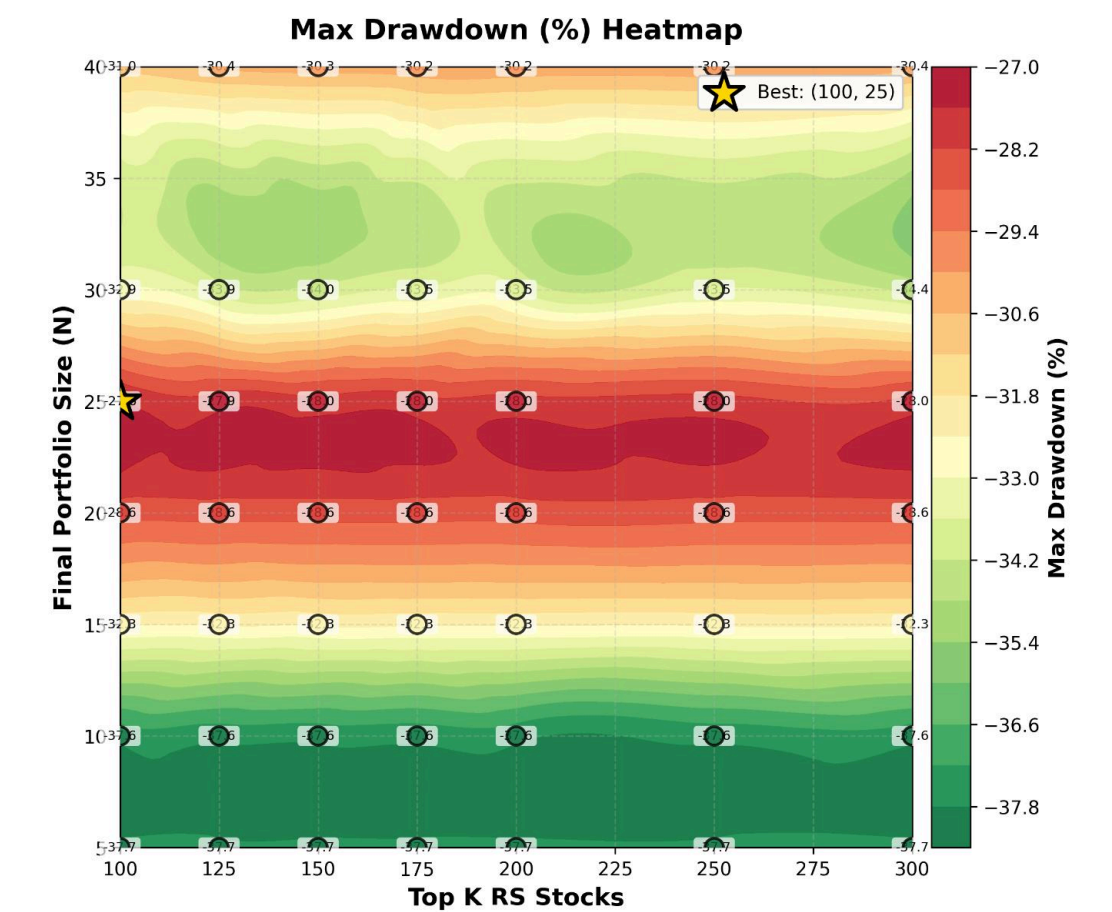
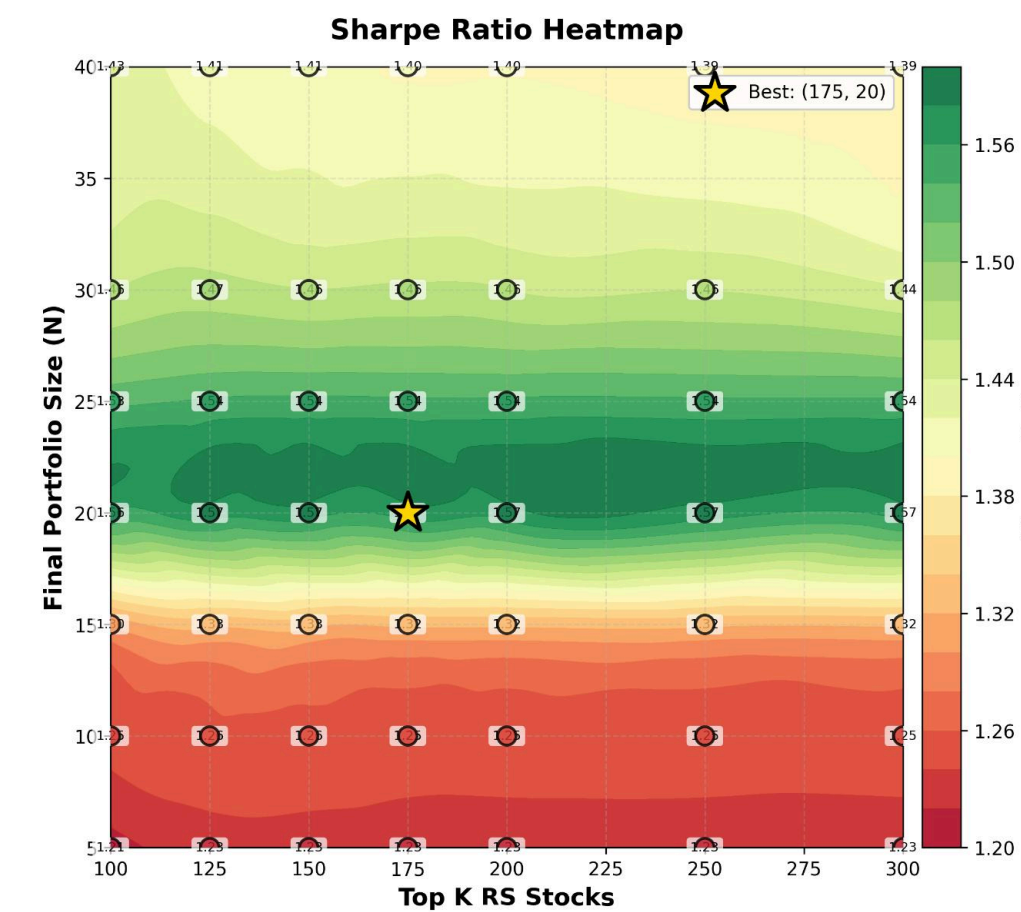
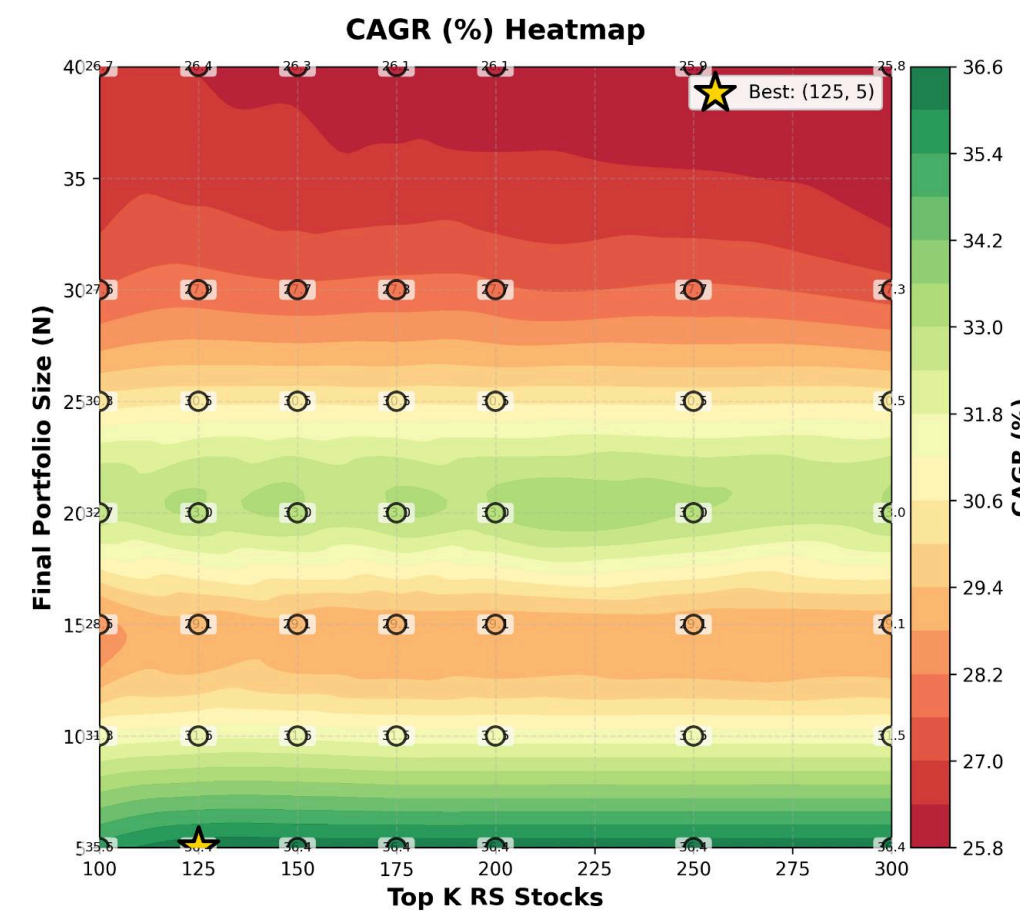
• Why This Strategy?

- 01 Macro-level sector regime awareness via Relative Strength pre-filter
- 02 Stock-level quality selection via residual momentum
- 03 Optimal weight allocation via Max Sharpe MVO
- 04 Dynamic responsiveness via dual rebalancing triggers
- 05 Position-level risk control via stop-loss

Hyperparameter Selection via Grid Search

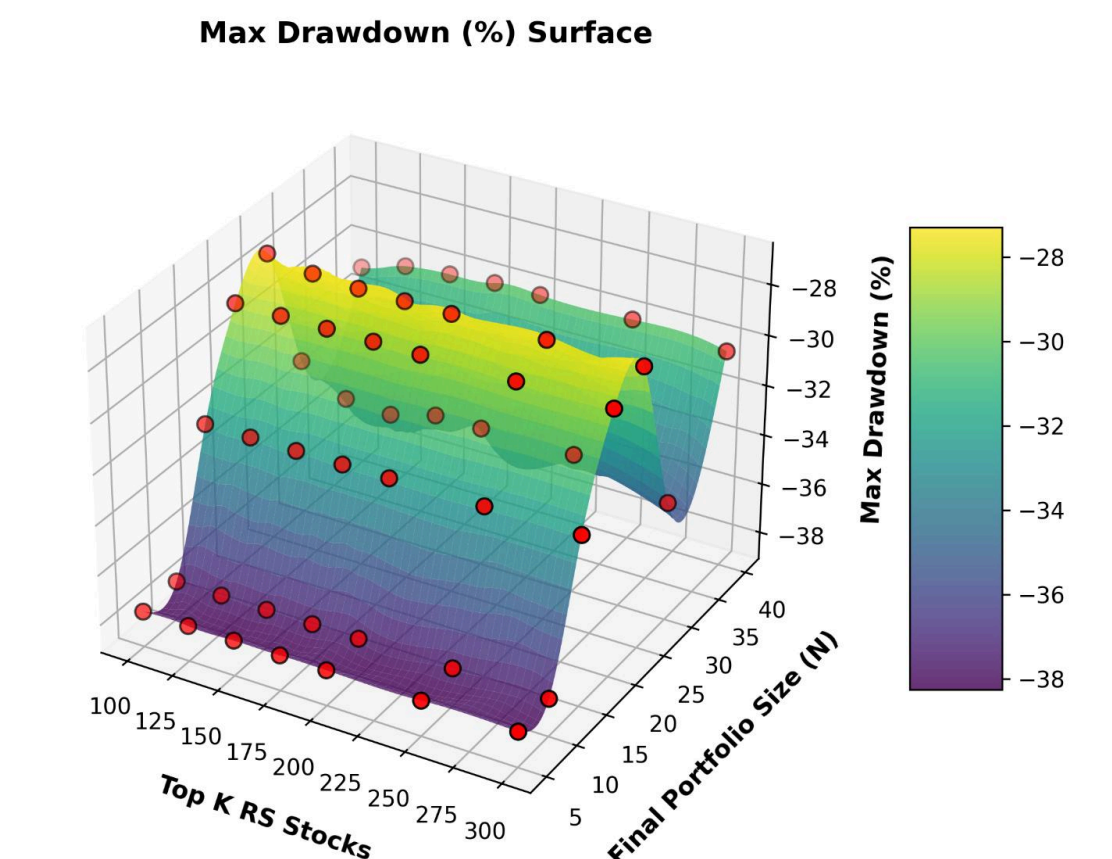
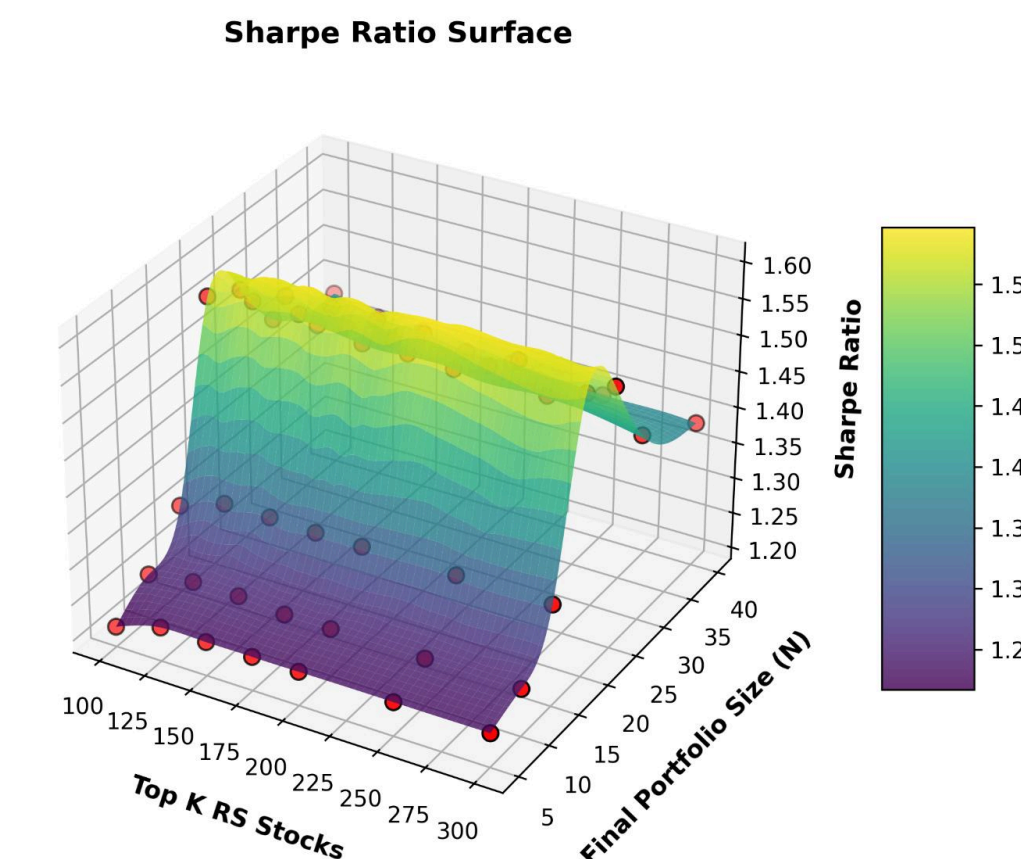
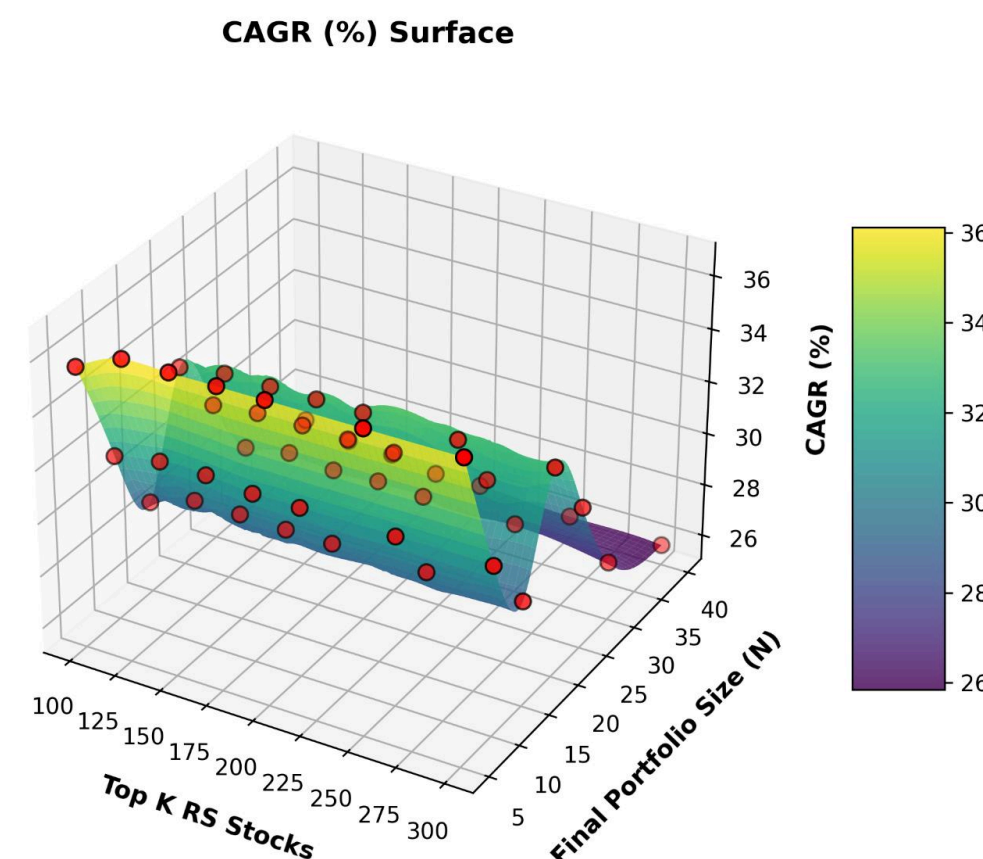
Explanation

- **K** = Number of stocks filtered out from the Universe using RS.
 - **N** = Final number of selected stocks using log-linear regression.
- These two parameters are not randomly chosen. Infact, they were selected via exhaustive **grid search** over $K \in \{100...300\}$ and $N \in \{5...40\}$.



Key Structural Findings:

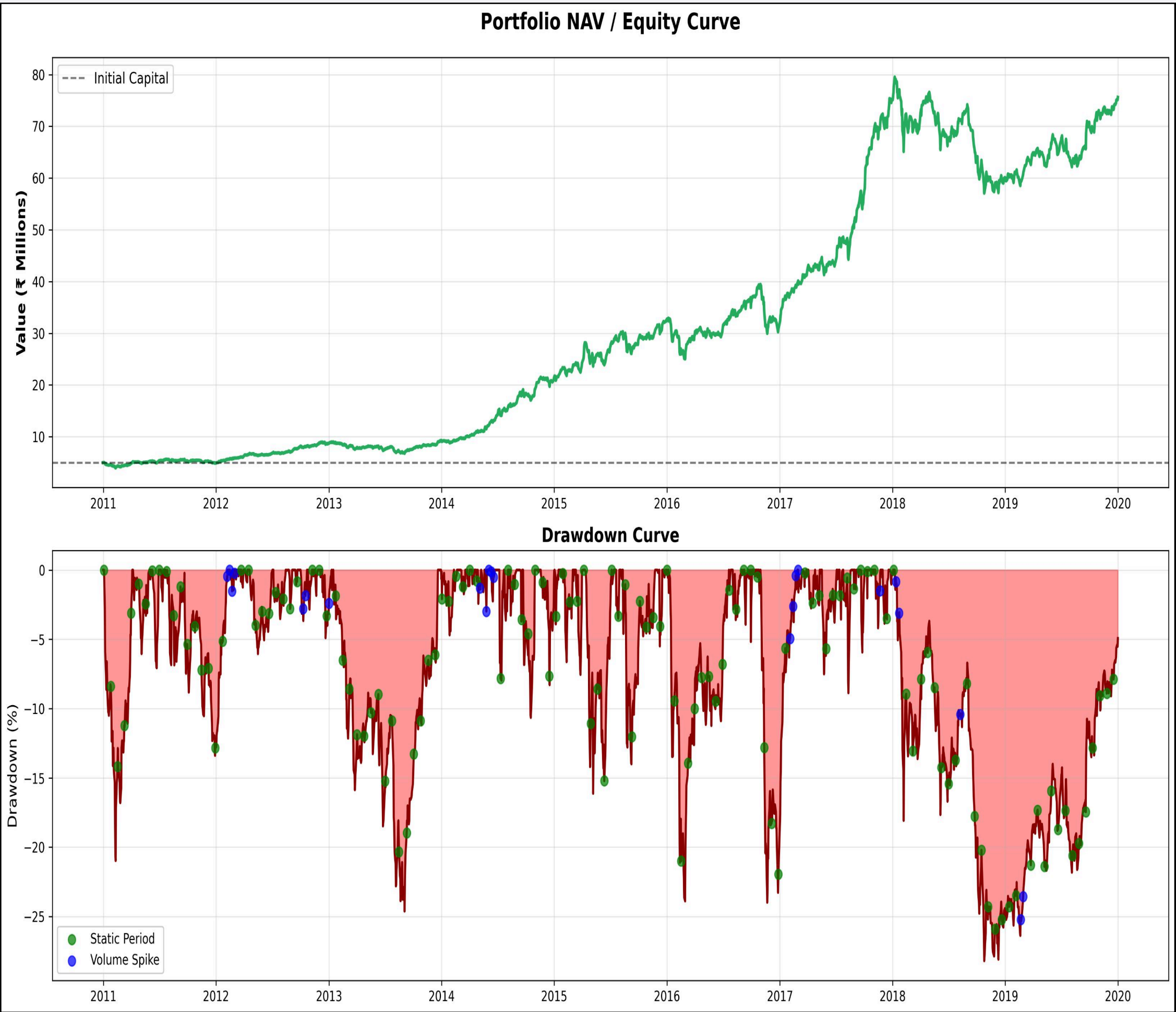
- CAGR and Sharpe most sensitive to N; sweet spot at $K = 200$, $N = 15-25$
- All three metrics nearly independent of K.
- Chosen: ($K = 200$, $N = 20$) — within high-Sharpe region, maintaining a balance between CAGR and Drawdown.



Results and Key Metrics

Metric	Strategy	Nifty 500	Edge
CAGR	37.51%	8.28%	+29.23%
Sharpe	1.59	0.60	+0.99%
Max DD	-26.96%	-28.65%	+1.69%
Up Capture	112.54%	100%	+12.54%
Down Capture	85.85%	100%	+14.15%
Calmar Ratio	1.39	0.29	+1.10

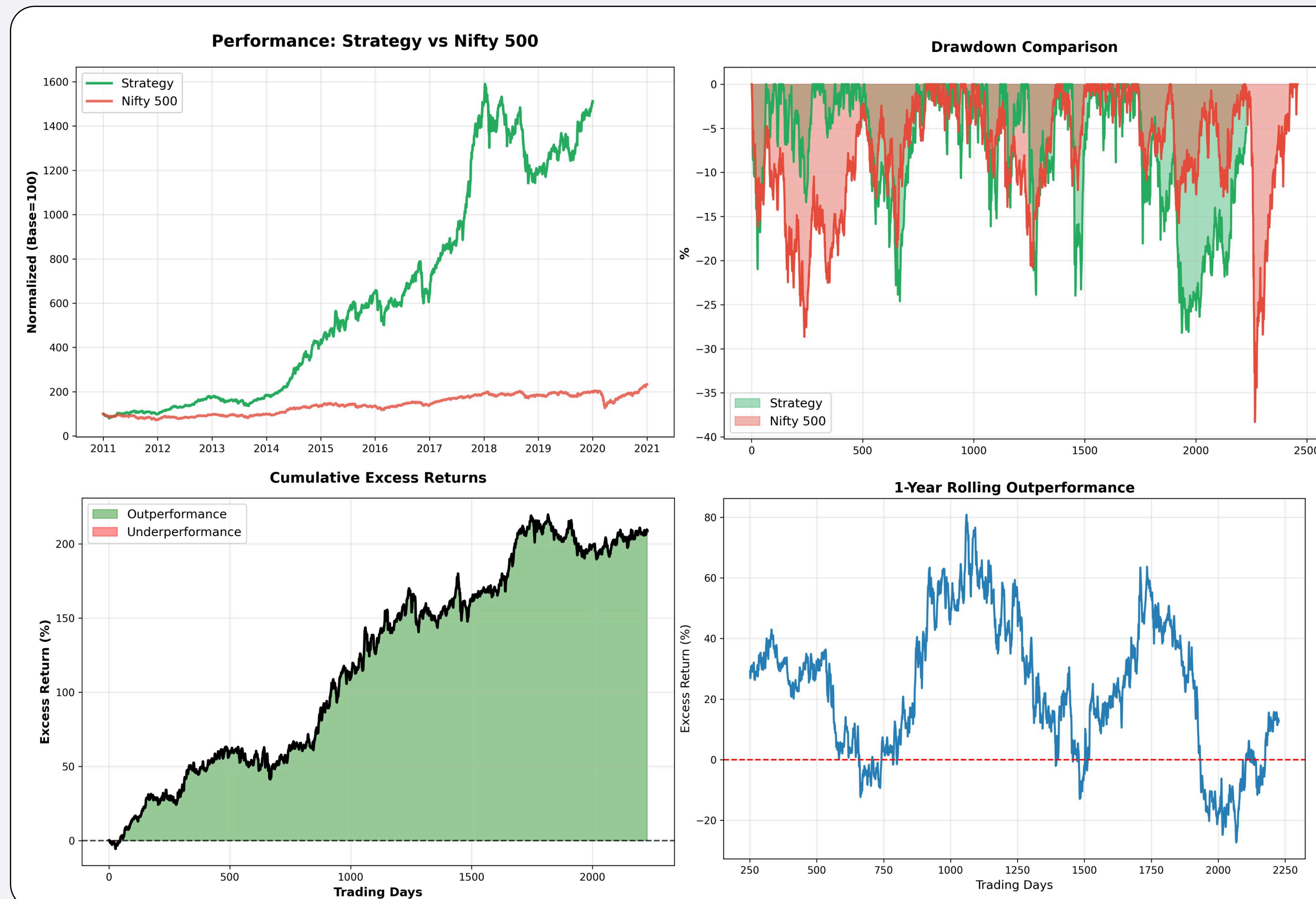
57.73% Win Rate
44.47% Avg. Turnover/Rebalance
1.511 Information Ratio



Rolling Outperformance vs Nifty 500

Window	Avg Outperformance	Worst Case
1 Year	+35.53%	-17.43%
3 Years	+201.14%	+38.21%
5 Years	+583.16%	+176.69%

- **1-year rolling window:** The only period with under-performance vs Nifty 500, indicating possible short-term lag.
- **3-year rolling window:** The strategy never under-performed the Nifty 500.
- **5-year rolling window:** Even in the worst case, it delivered nearly 3× the benchmark return.



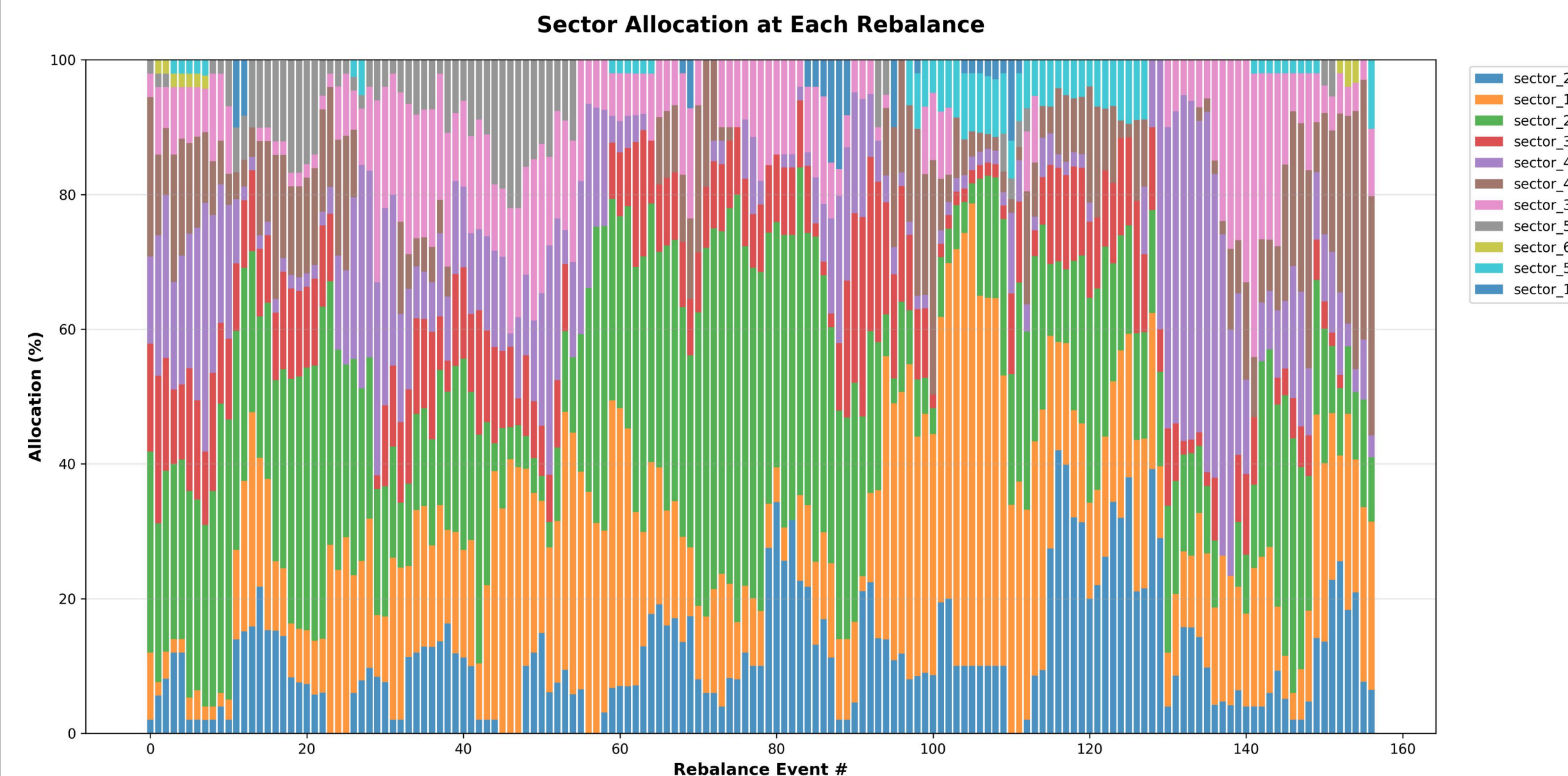
Risk Management, Diversification and Transaction costs

Risk Management

- **30% stop-loss** per position triggered 61 times, helping limit downside risk.
- **Volume-spike** rebalancing mostly occurred during market stress periods.
- **Calmar Ratio:** 1.39 vs Nifty's 0.39, showing the strategy outperforms Nifty in both CAGR and drawdown control.

Transaction Cost Reality

- **Avg turnover:** 44.47% per rebalance (65% lower than Approach 3's 128.21%).
- **Annualised transaction cost:** ~1.5–2.0%, already included in the 37.51% CAGR.
- **Net result:** Portfolio grew from ₹5M to ₹83.5M after all costs.



- **Weight bounds (2%–10%)** ensure no single stock dominates the portfolio, preventing concentration risk.
- The chart shows consistent **sector diversification** across all 157 rebalance events.
- Sector allocations dynamically adjust over time, maintaining balanced exposure while adapting to market conditions.

Thank you.

"Don't just hold the market.
Systematically outperform it."